

Segment-Level Non-Performing Assets Risk Prediction

Using a Random Forest Classification Model to forecast loan-segment NPA deterioration risk across HDFC Bank's SME, Education, Agriculture, Home, and Personal portfolios.

5
Loan Segments

100
Quarterly Records

7
Predictive Features

0.913
AUC-ROC Score

Python 3.x

scikit-learn

Random Forest

Power BI

FY2019–FY2023

HDFC Bank

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01 Project Objective

Core Goal — Build a Random Forest model that predicts, one year ahead, which of HDFC Bank's five loan segments faces the highest probability of Gross NPA ratio deterioration — enabling proactive, data-driven credit risk management.

Context and Goals

Non-Performing Assets represent one of the most pressing threats to Indian banking stability. When borrowers default, NPAs erode profitability, reduce liquidity, and constrain fresh lending. HDFC Bank's diversified portfolio — SME, Education, Agriculture, Home, and Personal loans — requires segment-specific early-warning intelligence that traditional reactive monitoring cannot provide. This project addresses that gap using quarterly data from FY2019 to FY2023, shifting risk management from crisis response to proactive prevention.

- **Predictive Accuracy:** Classify quarters likely to see GNPA ratio rise ≥ 0.5 pp within the next four quarters.
- **Segment Prioritization:** Rank all five loan segments by deterioration probability for targeted resource allocation.
- **Feature Transparency:** Identify which internal and macroeconomic drivers most influence NPA outcomes.
- **Actionable Guidance:** Translate probability scores into tiered, segment-specific intervention strategies.
- **Reproducibility:** Package the full pipeline from raw CSV ingestion to Colab execution for any analyst to replicate.

02 Data Description and Methods

2.1 Data Sources

Three complementary datasets were synthesized from secondary public sources to build a complete view of both internal portfolio dynamics and external macroeconomic pressures:

Dataset	Contents	Source(s)
Master Segment Table	Quarterly aggregates per segment: Total Advances, GNPA, GNPA Ratio, Provision, Net NPA, Recovery (100 rows $\times$ 15 cols)	Kaggle / public bank disclosures
Macro-Level Table	Quarterly indicators: Repo Rate, CPI, GDP YoY, Unemployment Rate, Agri Output Index, Industrial Growth	World Bank, FRED, RBI Bulletin, MoSPI
Loan-Level Table	Individual records: Loan Amount, Tenure, Interest Rate, Default Status, Recovery Amount, Slippages	Kaggle / Razorpay Historical Reports

2.2 Preprocessing Steps

Step	Action	Tool
1. Date conversion	QUARTER-END-DATE parsed to datetime (dayfirst=True, errors='coerce')	pandas.to_datetime
2. Symbol stripping	Remove '%' and ',' from numeric columns; cast to float	str.replace + pd.to_numeric
3. Sort & reset	Sort by LOAN TYPE then QUARTER-END-DATE for temporal integrity	df.sort_values
4. Target variable	GNPA_MAX_NEXT_4Q via groupby shift(-4) + rolling(4).max(); binary label ≥ 0.5 pp	pandas groupby
5. Lag features	GNPA_RATIO_lag1, lag2; REPO RATES_lag1; CPI_lag1; GDP_lag1	groupby.shift(1/2)
6. Rolling feature	TOTAL_ADVANCES_rollmean4 = 4-quarter rolling mean of advances	shift(1).rolling(4)
7. Growth feature	ADV_QOQ_GROWTH = quarter-on-quarter % change in total advances	pct_change()
8. Imputation	Median imputation on lagged/rolling NaN values (train-fit, test-transform)	SimpleImputer
9. Train/Test split	Time-based: before last 5 unique quarters = train (60); last 5 = test (25)	Boolean date filter

2.3 Modeling Approach & Evaluation Metrics

Algorithm: RandomForestClassifier — selected for its ability to handle non-linear relationships, robustness to NPA data outliers, and built-in feature importance. **Config:** n_estimators=200, max_depth=5, class_weight='balanced_subsample', random_state=42. Balanced weights correct for the 76:19 class imbalance (deterioration vs. stable).

- **Target (TARGET_DETERIORATION):** Binary — 1 if GNPA ratio rises ≥ 0.5 pp in next 4 quarters; 0 otherwise.
- **7 input features:** GNPA_RATIO_lag1, lag2, TOTAL_ADVANCES_rollmean4, ADV_QOQ_GROWTH, REPO_RATES_lag1, CPI_lag1, GDP_lag1.
- **Validation:** Strict time-based split — model never sees future data during training.

Metric	Score	Interpretation	Benchmark
AUC-ROC	0.9130	Discriminative ability across all thresholds	≥ 0.90 = Outstanding
Precision	0.9545	95.45% of flagged deteriorations were correct	≥ 0.80 = Excellent
Recall	0.9130	91.30% of actual deteriorations were caught	≥ 0.80 = Excellent
F1-Score	0.9333	Harmonic mean of precision and recall	≥ 0.80 = Excellent
Accuracy	88.0%	Overall proportion of correct predictions	$\geq 85\%$ = Excellent

03 Exploratory Data Analysis (EDA) Interpretation

3.1 Portfolio-Level Key Insights

- **Portfolio scale:** Average total advances = ■12.3M/quarter. Total provisions = ■76,63,361; NNPA = ■28,34,394; Total defaults = 259.
- **Systemic stress:** Aggregate GNPA ratio = 18.82% — nearly 1 in 5 loans was non-performing across the study period.
- **Macro shock impact:** FY2020 GDP contraction triggered synchronized GNPA spikes across SME, Home Loan, and Agriculture segments.
- **Class imbalance:** 76 deterioration vs. 19 stable labels (4:1); balanced class weights applied to prevent model bias.
- **Loan-level signal:** Defaulted loans had shorter tenure (45 vs. 57 months) and higher interest rates (9.68% vs. 9.46%), suggesting elevated EMI pressure correlates with default.

3.2 Segment Performance Benchmark (FY2019–FY2023)

Loan Segment	Avg Total Advances	Avg Gross NPA	GNPA Ratio	NPA Ratio	Profile
SME	■17,210,334	■5,855,161	33.76%	9.54%	Critical
Personal Loan	■8,361,284	■1,832,659	24.65%	6.99%	High
Agriculture	■5,962,361	■868,775	14.32%	4.12%	High
Education Loan	■5,664,407	■677,389	11.36%	3.35%	Moderate
Home Loan	■24,317,139	■2,377,429	9.99%	2.85%	Lowest
Portfolio Total	■12,303,105	■2,322,283	18.82%	5.37%	—

3.3 Visual Highlights

Figure	Type	Insight
Fig. 1	Power BI KPI Cards	■76,63,361 provisions; 259 defaults; 18.82% portfolio GNPA — systemic credit stress at a glance
Fig. 2	Multi-line GNPA + Recovery Chart	SME and Personal dominate GNPA contribution; recoveries are reactive, not preventive
Fig. 3	Macro vs. GNPA Overlay	FY20 GDP dip directly preceded portfolio-wide GNPA spike — strong macro sensitivity confirmed
Fig. 4a–e	Segment Line Charts	Agriculture: 47.43% GNPA spike Q3 FY19; SME: 52%+ in Q2 FY22; Home Loan: most stable base
Fig. 5	Horizontal Risk Bar Chart	SME = 85.09% highest deterioration risk; Personal Loan = 49.29% lowest
Fig. 6	Feature Importance Bar	ADV_QOQ_GROWTH tops at 0.2363; GDP_lag1 lowest at 0.0416
Fig. 7	Confusion Matrix Heatmap	21 true positives, 1 true negative; only 2 missed deteriorations in 25-record test
Fig. 8	Violin Plots × 5	SME shows widest GNPA distribution; Home Loan most concentrated — clear segment separation
Fig. 9	Annotated Time-Series	Per-segment GNPA trend with predicted risk % annotated at the most recent data point

3.4 Data Quality Concerns

- **Small dataset:** Only 100 quarterly rows (≈20 per segment) — test set confidence intervals are wide.
- **Secondary / simulated data:** Compiled from Kaggle and public sources, not HDFC Bank's internal systems. Results are indicative.
- **Post-FY2023 blind spot:** The model cannot account for RBI regulatory updates or portfolio shifts after the study period without retraining.
- **Segment-only granularity:** Predictions apply to entire loan segments, not individual borrowers.

04 Prediction Output

4.1 Predicted NPA Deterioration Risk by Segment

Applied to the most recent available quarterly snapshot per segment. Probability indicates the likelihood of GNPA ratio rising ≥ 0.5 pp within four quarters:

SME		85.09%
Education Loan		67.03%
Home Loan		56.25%
Agriculture		52.21%
Personal Loan		49.29%

4.2 Confidence Tiers & Recommended Actions

Risk Tier	Range	Segments	Recommended Action
Critical	$\geq 80\%$	SME (85.09%)	Immediate: monthly reviews, restructuring, dedicated risk team
High	60%–79%	Education Loan (67.03%)	Enhanced: track employment, income-based repayment options
Moderate	50%–59%	Home Loan (56.25%), Agriculture (52.21%)	Vigilance: seasonal monitoring, rate stress testing
Watch List	$< 50\%$	Personal Loan (49.29%)	Standard: tiered early-warning system with payment-pattern alerts

Interpretation Note — Probabilities near 50% indicate genuine model uncertainty, not safety. All predictions must be refreshed quarterly as new data arrives. Low confidence does not mean low risk — it means the pattern is less established in the training data.

05 Results, Interpretations & Recommendations

5.1 Model Performance Summary

Evaluated on a 25-record held-out test set (last 5 unique quarter-end dates). Confusion matrix: 21 true positives, 1 true negative, 2 false negatives, 1 false positive.

Outstanding Result — AUC-ROC 0.9130 (outstanding tier) · Precision 0.9545 · Recall 0.9130 · F1 0.9333 · Accuracy 88%. All metrics substantially exceed the 0.80 industry benchmark for credit risk classification, confirming the Random Forest successfully captured underlying NPA deterioration signals from the historical data.

5.2 Feature Importance — What Drives NPA Risk

Rank	Feature	Importance	Business Interpretation
1	ADV_QOQ_GROWTH	0.2363	Aggressive QoQ advance growth signals relaxed underwriting → future defaults
2	GNPA_RATIO_lag2	0.2035	NPA from 2 quarters ago; deterioration builds gradually, not suddenly
3	TOTAL_ADVANCES_rollmean4	0.1791	Structural portfolio expansion stretches monitoring capacity over time
4	GNPA_RATIO_lag1	0.1592	Most recent NPA level — clearest near-term signal of continuing stress
5	CPI_lag1	0.1270	Inflation erodes real borrower income; precedes repayment stress by ~1 quarter
6	REPO_RATES_lag1	0.0533	Monetary tightening raises EMI burdens, especially for variable-rate home loans
7	GDP_lag1	0.0416	Economic slowdown weakens income; SME segment most acutely sensitive

5.3 Segment-by-Segment Recommendations

SME — Critical Risk (85.09%)

SMEs lack diversified revenue, making them acutely sensitive to economic shocks. **Actions:** Monthly portfolio reviews · Dedicated SME risk team · Mandatory 6-monthly financial updates · Proactive restructuring at first sign of revenue stress · Strengthened collateral requirements for new disbursements.

Education Loan — High Risk (67.03%)

Risk driven by uncertain post-graduation employment. **Actions:** Track borrower employment status after graduation · Collaborate with placement cells · Introduce income-based repayment options · Regular co-signer financial health assessment.

Home Loan — Moderate Risk (56.25%)

Despite collateral backing, rising repo rates affect repayment capacity. **Actions:** Loan modification programs for income-stressed borrowers · Stress-test variable-rate mortgages under 1–2% rate hike scenarios · Real estate market trend monitoring.

Agriculture — Moderate Risk (52.21%)

Highly seasonal and weather-dependent segment. **Actions:** Seasonal monitoring aligned with kharif/rabi cycles · Commodity price and rainfall tracking · Flexible moratorium-style repayment structures · Mandatory crop insurance verification.

Personal Loan — Watch List (49.29%)

Lowest risk but fully unsecured. **Actions:** Strengthen income verification at origination · Tiered monitoring by loan size · Early warning systems flagging declining EMI regularity before formal default.

5.4 Potential Improvements

- **Larger dataset:** 10+ years of data across more banks would improve generalization and reduce small-sample risk.
- **SHAP explainability:** Add SHapley Additive exPlanations for per-prediction audit-ready breakdowns.
- **Hyperparameter optimization:** Apply Optuna or GridSearchCV for further AUC gains.
- **Individual borrower scoring:** Integrating loan-level data would enable account-level risk scores.
- **Real-time pipeline:** Migrate from batch quarterly scoring to a live Airflow-scheduled pipeline.

06 How to Reproduce

6.1 Environment Setup

Tested on Google Colab (Python 3.10). No GPU required.

```
# Recommended – Google Colab (zero local setup) # Open notebook:  
colab.research.google.com/drive/1TAiRILxTt15DMLakCbxS50KgiFpDPAqw # Local environment pip install  
pandas numpy matplotlib seaborn scikit-learn
```

6.2 Step-by-Step Execution

- | | |
|---------------|--|
| Step 1 | Upload RSK project data.csv to /content/ in Colab, or point the path variable to your local Kaggle dataset. |
| Step 2 | Run Steps 1–5 sequentially: imports → data load & clean → target variable → feature engineering → train/test split with median imputation. |
| Step 3 | Execute Step 6 to train the Random Forest (completes in under 30 seconds on CPU). |
| Step 4 | Run Step 7 for all evaluation metrics (AUC-ROC, Precision, Recall, F1, Confusion Matrix heatmap) and Step 8 for feature importance chart. |
| Step 5 | Run Step 9 to generate final segment risk probabilities and horizontal bar chart. Steps 10–11 produce per-segment annotated time-series visuals. |
| Step 6 | For Power BI: import Master, Macro, and Loan tables; build star schema with Date Table as hub; use DAX for GNPA KPIs and segment filters. |

07 Deployment / Usage Notes

7.1 Interaction Modes

- **Colab Notebook (primary):** Upload updated quarterly CSV → re-run all cells → inspect the risk bar chart and annotated time-series. Under 2 minutes end-to-end.
- **Batch quarterly scoring:** Append new quarter row to master CSV, re-run Step 9, share updated probability table with the risk committee.
- **Power BI dashboard:** Refresh the data model each quarter to update KPI cards, trend charts, and macro-correlation visuals for monthly risk reviews.
- **Standalone Python script:** Export notebook as .py and schedule via cron or Airflow for automated quarterly scoring.

7.2 Quarterly Deployment Checklist

Before Each Run	After Results Generated
■ Verify column schema matches training layout	■ Escalate any segment crossing 70% threshold
■ Re-run imputer only if full model retraining	■ Update Power BI dashboard with latest output

08 Limitations & Ethical Considerations

8.1 Known Biases and Data Limitations

- **Secondary data:** Dataset compiled from Kaggle and public sources — not HDFC Bank's proprietary systems. Real-world magnitudes may differ.
- **Small sample:** ≈20 observations per segment; high performance metrics may partially reflect fitting to a small, well-behaved dataset.
- **Historical bias:** Training on FY2019–2023 embeds pandemic-era anomalies (GDP crash, RBI moratorium) that may not recur.
- **Concept drift:** Structural changes in lending regulations or portfolio strategy post-FY23 may invalidate learned feature relationships.

8.2 Responsible Use & Safeguards

- **Human-in-the-loop mandatory:** All high-risk segment flags must be reviewed by a qualified credit risk officer before triggering borrower-facing action.
- **No individual scoring:** This model does not score individual borrowers. Using segment-level risk to deny individual loans is ethically inappropriate.
- **Transparent limitations:** Predictions must always be shared with their confidence context and the secondary-data caveat.
- **Regular retraining:** Retrain quarterly as new data arrives. Monitor Population Stability Index (PSI) on GNPA_RATIO_lag1 for drift.
- **Agricultural sensitivity:** Farmers face uncontrollable external shocks — interventions must use empathetic restructuring, not punitive collection.

09 Appendix — Glossary & Key Figures

9.1 Glossary of Key Terms

Term	Definition
NPA	Non-Performing Asset — loan where borrower has not made payments for 90+ days.
GNPA	Gross NPA — total value of NPAs before deducting provisions held by the bank.
GNPA Ratio	Gross NPA as % of total advances; primary measure of portfolio credit stress.
TARGET_DETERIORATION	Binary label: 1 if GNPA ratio rises ≥ 0.5 pp in next 4 quarters; 0 otherwise.
ADV_QOQ_GROWTH	Quarter-on-quarter % change in total advances; top-ranked model predictor (0.2363).
GNPA_RATIO_lag1/2	GNPA ratio from 1 or 2 quarters prior; captures NPA momentum over time.
TOTAL_ADVANCES_rollmean4	4-quarter rolling mean of total advances; smooths short-term noise.
CPI_lag1	Consumer Price Index from the previous quarter; proxy for inflation pressure.
Repo Rate	RBI's benchmark lending rate; changes directly affect variable-rate EMIs.
AUC-ROC	Area Under ROC Curve; measures discriminative ability across all thresholds.
Balanced Class Weight	Up-weights minority class to correct for the 76:19 label imbalance.
Time-Based Split	Train/test divided by date boundary (not random) to prevent data leakage.
PSI	Population Stability Index — monitors feature distribution drift between quarters.

9.2 Key Figures & Tables Reference

Reference	Title	Location
Fig. 1	Portfolio KPI Dashboard	Power BI — Page 1 Overview
Fig. 2	Segment GNPA & Recovery vs. Quarter	Power BI — Segment Analysis
Fig. 3	Macro-Economics Influence on GNPA	Power BI — Macro Correlation
Fig. 4	Per-Segment Trend Lines (5 charts)	Power BI — Segment Deep-Dives
Fig. 5	Predicted Risk of NPA Deterioration — Bar Chart	Notebook Step 9 / Chapter 7
Fig. 6	Feature Importances — Bar Chart	Notebook Step 8 / Chapter 6.5
Fig. 7	Confusion Matrix Heatmap	Notebook Step 7 / Chapter 6.4
Fig. 8	Violin Plots — Top 5 Features by Loan Type	Notebook Step 9 (violin section)
Fig. 9	Annotated GNPA Time-Series per Segment	Notebook Step 10
Fig. 10	Risk Factor Analysis: SME vs. Average	Notebook Step 10 (highest risk)
Fig. 11	Risk Factor Analysis: Personal Loan vs. Average	Notebook Step 10 (lowest risk)
Table A	Segment Performance Benchmark	Chapter 5.3.3 / Appendix 8.6.2
Table B	Defaulted vs. Non-Defaulted Loan Characteristics	Chapter 5.3.6 / Loan-level data
Table C	Power BI Data Model (Star Schema)	Chapter 4 / Appendix 8.6.3